CHENG-HSUAN LEE

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Legged robotics | Actuator design | Learning-based control | Sim-to-real experiments

EDUCATION

National Taiwan University | B.S. in Mechanical Engineering

Taipei, Taiwan | 2023-2027 (expected)

Selected coursework: Microprocessor Controlled Systems; Automatic Control; Applied Electronics; Mechanism; Machine Design Theory; Engineering Graphics; Practice of Mechanical Engineering; Manufacturing Processes; Workshop Practice; Mechanical Engineering Laboratory I/II; Dynamics; Numerical Analysis.

Taipei Municipal Chien Kuo High School | 2020-2023 | CKHS Robotics Team

PUBLICATION AND CONFERENCE PRESENTATION

Cheng-Hsuan Lee, Yi-Cheng Wang, Yu-Hsuan Chi, Yu-Mei Huang, Yu-Cheng Kuo, and Chung-Hsien Kuo. **"Sim-to-Real Performance Analysis of a Custom Quadruped Robot Using Gait-Generator-Guided Reinforcement Learning."** *2026 International Automatic Control Conference (CACS 2026)*, IEEE conference proceedings, Taichung, Taiwan, October 21-24, 2026. [Paper]

- **First author; in-person English oral presenter.** Personally presented the research at CACS 2026.
- **Best Student Paper Award - Honorable Mention;** selected among **11 finalists** before the final award evaluation.

RESEARCH AND LABORATORY EXPERIENCE

NTUME ASR Lab | Learning-Based Quadruped Robot Integration

Undergraduate researcher | Control Systems Laboratory | Rifle custom 12-DoF quadruped

Research across **actuator hardware, embedded motor control, locomotion learning, and physical validation**. Connected mechanical behavior and controller feedback to the sim-to-real investigation reported in our first-author CACS paper.

- **Actuator and reducer development:** designed and fabricated cycloidal and planetary reduction-gearbox concepts; examined packaging, backlash, bearing support, shaft strength, assembly tolerance, and serviceability for repeated robot tests.
- **Motor-control interfaces:** worked on communication packets, Arduino/Teensy-side programming, command/readback routines, calibration, and position/velocity/current feedback with ODrive-controlled actuation.
- **Learning experiments:** ran and documented Isaac Lab / Isaac Sim locomotion workflows using PPO and imitation learning; compared clearance, posture, contact, torque, and action-smoothness terms using checkpoints, TensorBoard records, and simulation videos.
- **Hardware-feasible gait guidance:** the study first checked a robot-specific gait reference by direct hardware execution, then used joint- and foot-tracking references to guide PPO toward physically executable locomotion.
- **Deployment and failure diagnosis:** used ROS 2 launches, rosbag data, joint feedback, IMU records, current traces, and physical inspection to separate policy errors from sensing, communication, calibration, and drivetrain faults.
- **Quantitative transfer evidence:** the study evaluated two guided controllers over **six hardware recordings**; all sustained walking at a **0.4 m/s straight, level-ground command**, retaining **84.9-90.0%** of simulated swing amplitude with **0.980-0.990** phase correlation.
- **Failure analysis and next experiments:** traced the recorded pure-RL deployment failure to inadequate foot motion and actuator loading; documented residual joint-tracking, body-motion, and current gaps to motivate hardware identification and better feedback estimation.

SELECTED ENGINEERING PROJECTS

Self-Built Simple Robot Dog | Team Lead, Mechanical Designer, Controls Developer

Microprocessor Controlled Systems | Course-built eight-servo platform, distinct from the ASR Lab research robot

- **Mechanical design and integration:** led an eight-servo quadruped build; designed body and leg modules around servo placement, wire routing, battery balance, and repair access, using a serviceable 3D-printed structure for repeated gait tests.
- **Gait-to-hardware implementation:** translated MATLAB trajectory and inverse-kinematics reasoning into ESP32/PCA9685 control; calibrated servo directions and neutral angles, separated gait timing from servo output, and demonstrated walking, standing, sitting, waving, and rocking **(A+)**.

Quadcopter Ball-Transfer Drone | Team Lead and Systems Integrator

Practice of Mechanical Engineering | Full marks and A+ course grade

- **Mission-level system design:** led integration of the airframe, ball-transfer payload, propulsion, power, and flight electronics; balanced payload placement and flight stability while coordinating subsystem tests and final-run preparation.
- **Testing and debugging:** used thrust-to-throttle measurements, SBUS signal checks, and vibration/drift diagnosis to isolate propulsion, controller, and structural issues. Earned **full marks / A+**; many peer teams referenced our architecture and testing approach.

Foldable Ball Transfer Mechanism | Team Lead

Machine Design Theory | Packaging-constrained deployable mechanism

- **Packaging-driven mechanism design:** led concept selection and folding architecture to reconcile compact storage with deployed reach, structural stiffness, and ball-transfer motion.
- **Fabrication and validation:** combined hinges, latch hardware, MDF, and 3D-printed parts; evaluated locking, assembly sequence, tolerance, and repair access through repeated builds, achieving reliable deployment and ball transfer in the final test.

LEADERSHIP AND TEAM EXECUTION

- **Engineering project leadership:** led the robot dog, drone, and foldable-mechanism teams, translating open-ended goals into design decisions, task ownership, integration tests, and final deliverables.
- **CKHS Robotics Team / FIRST Robotics Competition:** served as programming, strategy, and scouting leader; connected autonomous logic, driver needs, mechanism limits, match data, and repair constraints during regional and championship competition.
- **Evidence-based coordination:** supported mobile scouting tools and Excel analysis workflows for robot comparison, alliance selection, and match decisions; coordinated mechanical, electrical, and software troubleshooting under deadlines.

AWARDS AND RECOGNITION

- **2026 CACS:** Best Student Paper Award - Honorable Mention; one of 11 finalists; first author.
- **2022 FRC Sacramento Regional:** Finalist and Industrial Design Award.
- **2022 FRC New Taipei City x Hon Hai Regional:** Engineering Inspiration Award; qualified for Houston Championship.
- **2023 FRC Los Angeles and San Diego Regionals:** Excellence in Engineering Award at each regional.
- **2nd ISS Kibo Robot Programming Challenge, Taiwan Preliminary:** Merit Award.

TECHNICAL SKILLS

Mechanical: CAD, engineering drawings, 3D printing, machining, mechanism design, cycloidal/planetary reducers, actuator packaging, and tolerance/assembly planning.

Reinforcement Learning (RL): PPO/DWAQ workflows, gait-reference-guided learning, imitation learning, reward shaping, observation/action mapping, and sim-to-real policy evaluation in Isaac Lab / Isaac Sim.

Robotics and control: ROS 2, rosbag, TensorBoard, URDF, Pinocchio, MATLAB/Simulink, and motor calibration.

Embedded and software: Python, C++, Arduino, ESP32, Teensy, ODrive, PCA9685, RS485, UART, I2C, PWM, Linux, and Git/GitHub.

Experiments: motor feedback analysis, servo calibration, thrust testing, signal-chain debugging, failure isolation, and technical reporting.